Aryabrata Basu

Research Statement

"Every great advance in science has issued from a new audacity of imagination." - John Dewey, The Quest for Certainty, 1929

Origins

In 1993, as a third grader, I visited my uncle Debabrata, who had promised to show me a personal computer powered by Intel's original Pentium processor. What fascinated me most was not the specifications but the interactive magic of DOS games—how a few keystrokes could move virtual characters, vehicles, and worlds in real time. That experience sparked a lasting fascination with computer graphics and human-computer interaction. Today, my work has evolved from those early encounters to developing immersive simulations of imagined and data-driven environments that enable rigorous analysis of human behavior, decision-making, and interaction across complex spatial and temporal contexts.

Motivation and Research Philosophy

Since Ivan Sutherland's vision of the "Ultimate Display" in 1968, virtual reality has progressed from experimental laboratory systems to widely accessible platforms. Yet fundamental challenges remain. Many users still encounter simulator sickness, ergonomic discomfort, and physical or cognitive strain. These limitations can erode trust in immersive systems and prevent their broader adoption, particularly in fields where accuracy, comfort, and sustained engagement are critical.

My motivation is to address these barriers through a combination of technical innovation and human-centered design. I develop immersive systems that minimize physical burden, adapt to user needs, and integrate seamlessly into both everyday decision-making and high-stakes professional contexts. This requires not only engineering efficiency in hardware and software but also designing interaction models that account for cognitive load, perceptual comfort, and user trust over time.

My research philosophy rests on three pillars. First, immersive technology must be intuitive and accessible so that expertise with the interface does not become a prerequisite for meaningful engagement. Second, it must be validated through rigorous, repeatable evaluations that account for diverse user backgrounds and contexts of use. Third, immersive environments should foster sustained interaction—whether for education, training, or collaboration—by aligning the design of virtual spaces with human cognitive strengths and limitations.

Ultimately, I view VR not as an isolated technology but as a platform that can bridge disciplines, translating complex information into forms that humans can intuitively explore, understand, and act upon.

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Previous Work

My doctoral research produced the **Ubiquitous Collaborative Activity Virtual Environment** (UCAVE), a lightweight smartphone-based VR framework supporting untethered, portable immersive experiences. Through systematic studies, I examined:

- The ergonomic benefits of untethered VR designs in improving performance.
- The influence of environmental fidelity, particularly outdoor realism, on task outcomes.
- The role of prior gaming experience and input device familiarity in shaping VR usability.
- The effect of head-mounted display deployment on user trajectories in spatial navigation tasks.

These contributions established guidelines for creating VR experiences that balance usability, accessibility, and immersion.

Current Research Endeavors

Building on these foundations, my current work expands into interdisciplinary applications of immersive technologies.

Persistent Virtual Reality Spaces
I am developing Persistent Virtual Reality Spaces (PVRS) that sustain engagement, learning, and collaboration over long durations. Drawing from Friedrich Froebel's educational philosophy, PVRS combines structured experiential learning with intuitive design. My recent TVCG journal paper, accepted at IEEE VR 2025, examines the long-term effects of novelty in VR learning, directly guiding PVRS design for comfort and retention.

Digital Cardiology and Advanced Medical Visualization
As PI on the NSF EPSCoR DART-funded *Digital Cardiology* project (USD 100,000), I am creating immersive conformal-mapped visualizations of cardiac arterial structures to improve diagnostic precision and pre-surgical planning. This work transforms static medical imaging into interactive decision-support tools.

Immersive Agriculture Decision Support Systems
In collaboration with the University of Arkansas, I am developing VR and AR tools that combine satellite-derived crop phenology data, climate variables, and management practices into predictive yield visualizations. Engaging stakeholders directly ensures these tools address practical challenges in Arkansas agriculture.

Secure Real-Time Distributed 3D Mesh Synchronization
I am building Unity-based protocols for distributing high-fidelity 3D meshes across networked environments with minimal latency. This work integrates custom serialization, adaptive chunking, and AES encryption to support secure industrial, educational, and collaborative VR.

Cybersecurity Training in Simulated Industrial Environments
As co-lead on a U.S. Department of Energy initiative (USD $5 Mil), I help design immersive simulations of industrial control systems for cybersecurity workforce development. These environments train participants in threat detection, incident response, and operational resilience.

Human Spatial Decision-Making in VR
I continue investigating spatial cognition in VR using neural networks and emerging transformer

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architectures to model user navigation and decision-making. This research supports adaptive and personalized training systems.

Future Research Goals

My long-term goal is to advance immersive systems that enhance cognitive performance, decision-making, and comfort in persistent virtual reality environments. Building on my work in PVRS, digital cardiology, and agriculture, I plan to integrate transformer-based models to predict and adapt to nuanced user behaviors in real time. These adaptive systems will personalize interaction, optimize engagement, and mitigate cybersickness.

A key element of this vision is the development of *DataEcho*, an open and extensible immersive analytics framework that unifies secure data integration, adaptive visualization, and real-time behavioral modeling. *DataEcho* will be designed to encourage collaboration, reproducibility, and community contributions, enabling researchers and practitioners to build upon its core capabilities. By supporting transparent and interoperable workflows, the platform will make it easier to translate complex data streams into actionable, context-aware insights across domains such as healthcare, agriculture, cybersecurity, and national defense.

In cybersecurity, I will extend my secure mesh synchronization framework into multi-user VR training platforms for industrial and critical infrastructure defense, combining immersive simulation with AI-driven threat detection and performance analytics. These same technologies have the potential to inform defense-related projects, such as mission rehearsal environments, situational awareness tools, and decision-support systems for complex operational scenarios.

In medical and agricultural contexts, I plan to develop immersive visualization pipelines that integrate live sensor streams, predictive analytics, and multi-scale spatial models—supporting intraoperative guidance in cardiology and real-time field decision-making in agriculture.

By combining machine learning, secure networked environments, and openly available domain-specific visualization tools, my research will deliver immersive systems that are adaptable, transparent, and impactful. These platforms will serve as experimental testbeds for understanding human behavior, as well as practical solutions for high-stakes decision-making in both civilian and defense applications.

Closing Statement

Computer Graphics and Human-Computer Interaction remain rich with challenging problems. I aim to address these through research that advances immersive visualization, strengthens human-computer interaction, and applies VR to domains where insight and precision are critical.

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